🏷️ CHEMISTRY CH.15 — MISCELLANEOUS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📑 Topics in this chapter:

1. Atoms, Molecules & Basic Definitions

- Element: Same number of protons wale atoms — Fe, Cu, Ag, Au, H, C, N, O. *[Q1611]*
- Molecule: 2+ atoms chemically bonded — O_2, N_2, NH_3, H_2O, CO_2. *[Q1683]*
  - Diatomic: O_2, N_2, H_2. Triatomic: CO_2. Polyatomic: NH_3, H_2O, S_8. *[Q1683]*
- Alloys: Red gold (Au+Cu), white gold (Au+Ni+Zn), sterling silver (Ag+Cu) — NOT pure molecules. *[Q1611]*
- Pure substance: Methane, copper, silver, gold, water, glucose — fixed composition. *[Q1653]*
- Panch Tatva: Air, Earth, Fire, Sky, Water — Acharya Kanada (Kashyap) ne paramanu (atom) propose kiya. *[Q1672]*
- Neon (Z=10): $1s^2\ 2s^2\ 2p^6$; K=2, L=8; 2 shells completely filled — noble gas. *[Q1629]*
- Electronic configurations: C = $1s^2 2s^2 2p^2$; Mg = $1s^2 2s^2 2p^6 3s^2$; Ar = $1s^2 2s^2 2p^6 3s^2 3p^6$. *[Q1629]*
- ⚠️ Trap: "Atom of a compound" — ye statement galat hai; compound = molecule/bonded atoms. *[Q1675]*
- ⚠️ Trap: Particles of matter stationary nahi — hamesha continuous motion mein hain. *[Q1677]*

2. States of Matter & Phase Changes

- Solid: Definite shape & volume, rigid, incompressible, strong intermolecular forces, least diffusion. *[Q1676, Q1677]*
- Liquid: Definite volume, no fixed shape, weaker forces than solids. *[Q1676]*
- Gas: No fixed shape/volume, weak forces, rapid diffusion, most intermolecular space. *[Q1676, Q1677]*
- Plasma: 4th state; free electrons + ions; stars (Sun) mein; >99% visible universe. *[Q1610, Q1618]*
- Bose-Einstein Condensate (BEC): 5th state; super cold atoms; S.N. Bose + Einstein. *[Q1524 from previous chapter]*
- Intermolecular forces: Solid > Liquid > Gas. *[Q1654]*
- Compressibility: Volume change under pressure — Gas > Liquid >> Solid. *[Q1662]*
- Rigidity: Fixed shape due to strong forces — solids. *[Q1659]*

- State change: Temperature change se hota hai — energy add/remove. *[Q1682]*
- Physical changes: Melting, freezing, liquefaction, vaporisation, shredding — reversible. *[Q1605]*
- Chemical changes: Burning, cooking, rusting, rotting, fermentation — new substance forms. *[Q1605]*
- Evaporation: Liquid → Gas; cooling effect — salt from seawater, water in earthen pot, acetone on palm. Rate increases with both surface area aur temperature (humidity/wind speed bhi factors hain — humidity badhne se rate ghatta hai). *[Q1495, Q1580, Q1605, Q1642]*
- Sublimation: Solid → Gas — Iodine, Camphor, Dry ice (solid CO_2), NH_4Cl, Anthracene, Naphthalene. *[Q1625, Q1652, Q1681]*
- Deposition: Gas → Solid directly (water vapour → ice). *[Q1580]*
- Condensation: Gas → Liquid — morning dew, foggy windows. *[Q1419, Q1580]*
- Liquid Nitrogen boiling point: -196°C (melting point -209.9°C); fractional distillation of liquid air se banta hai; low viscosity colourless coolant. *[Q1587]*
- ⚠️ Trap: "Solid CO_2 directly changes to liquid" — galat; ye directly gas mein jaata hai. *[Q1658]*
- ⚠️ Trap: Solids ka definite shape hota hai — indefinite nahi. *[Q1676]*
- ⚠️ Trap: Fermentation chemical change hai — physical nahi. *[Q1605]*

3. Mole Concept, Atomic Structure & Isotopes

- Mole: Amount of substance jo 12 g Carbon-12 mein atoms ke barabar elementary particles contain kare = 6.022×10^{23} (Avogadro's number, N_0). *[Q1617, Q1673, Q1678]*
- 1 mole kisi bhi particle mein hamesha 6.022×10^{23} particles hote hain — atomic/molecular mass pe depend nahi. *[Q1673]*
- 16 g Oxygen = 1 mole = 6.022×10^{23} atoms. *[Q1666]*
- 1 mole H_2O = 18 g (M = 2×1 + 16 = 18). *[Q1645]*
- $(NH_4)_2SO_4$ mein total atoms = 15 (H=8, N=2, O=4, S=1). *[Q1671]*
- Carbon-12 = atomic mass reference standard. *[Q1619]*
- Isotopes: Same atomic number, different mass number — H-1/2/3; C-12/13/14; O-16/17/18. *[Q1619, Q1632]*
- Isobars: Same mass number, different atomic number — Ca-40 (Z=20) & Ar-40 (Z=18). *[Q1632]*
- Isotones: Same neutrons — ^{36}S, ^{37}Cl, ^{38}Ar. *[Q1632]*
- Ionization energy unit: kJ mol^{-1}. *[Q1670]*
- Dalton's atomic theory: Matter = small particles called atoms; element, compound, mixture sab atoms se. *[Q1507, Q1518 from previous chapter]*
- Law of Conservation of Mass: Reactants mass = Products mass. *[Q1664]*
- Law of Definite Proportions: Compound mein elements fixed mass ratio mein — CO_2 (C:O = 3:8), H_2O (H:O = 1:8), NH_3 (N:H = 14:3). *[Q1669, Q1664]*

4. Chemical Bonding: Ionic vs Covalent

- Silicon vs Carbon: Same group (14); Si atom larger; Si-Si bond C-C bond se lamba aur kamzoor → silicon long-chain compounds hyper-reactive. *[Q1561]*
- Covalent bond: Electron sharing; metals + non-metals ke beech nahi. *[Q1566]*
- N_2: Triple bond — 6 shared electrons (3 pairs). *[Q1562, Q1577]*
- CO_2: O=C=O; carbon har oxygen ke saath double bond — 2 electrons share. *[Q1564]*
- Covalent compounds examples: NH_3, O_2, CH_4, H_2O, P_2O_5. *[Q1566, Q1574]*
- Ionic bond: Electron transfer; metals + non-metals. *[Q1569]*

- Ionic compounds examples: $NaCl$, $CaCl_2$, CaO, $MgCl_2$. $AgCl$ (Silver Chloride) bhi ionic compound hai — $Ag^+ + Cl^-$ combine hoke banta hai ($AgNO_3$ solution mein $NaCl$ daalne se white precipitate). *[Q1566, Q1571, Q1685]*
- Electrovalent = Ionic — same cheez. *[Q1569]*
- Anion: Negative ion — gains electrons ($F \rightarrow F^-$). *[Q1624]*
- Cation: Positive ion — loses electrons ($Na \rightarrow Na^+$). *[Q1624]*
- NH_4^+: Cation (+1). CO_3^{2-}, NO_3^-, OH^-: Anions. *[Q1674]*
- NaCl: Na^+ (cation) + Cl^- (anion); Na loses $1e^-$ (Neon config), Cl gains $1e^-$ (Argon config). *[Q1628]*
- Electronegativity: Atom's tendency to attract electrons; Left→Right increases, Top→Bottom decreases.
  - Top: F > O > N > Cl > Br. *[Q1637]*
- NaOH: Ionic (Na^+ & OH^-) + covalent (O-H bond) — dono bonding. *[Q1576]*
- Ionic compounds ke high melting/boiling points: Strong inter-ionic (electrostatic) attraction todne ke liye bahut zyada energy chahiye — isiliye melting/boiling points high hote hain. Examples: CaO (2850K) > NaCl (1074K) > $CaCl_2$ (1045K) — CaO sabse zyada boiling point wala. *[Q1563, Q1573, Q1578, Q1661]*
- ⚠ Trap: N_2 mein 6 shared electrons hain — 4 ya 8 nahi. *[Q1562]*
- ⚠ Trap: $MgCl_2$ ionic hai — covalent nahi. *[Q1566]*
- ⚠ Trap: NaOH pure covalent nahi — ionic + covalent dono hai. *[Q1576]*
- ⚠ Trap: Electron add karne se anion banta hai. *[Q1624]*
- ⚠ Trap: Intermolecular force solid > liquid > gas. *[Q1654]*

5. Laws of Chemical Combination & Equations

- Chemical equation symbols: + (multiple reactants/products), → (reactants to products), ⇌ (reversible), Δ (heated). *[Q1622]*
- State symbols: (s) = solid, (l) = liquid, (g) = gas, (aq) = aqueous. *[Q1622]*
- Balanced equation: Reactants aur products donon taraf har element ke atoms barabar. *[Q1081 from previous chapter]*
- Law of Conservation of Mass + Law of Definite Proportions — Dalton's atomic theory se explain hote hain. *[Q1664]*

6. Acids, Bases, Salts & Neutralization

- Alkali: Base jo water mein dissolve hoti hai — NaOH, KOH, $Ca(OH)_2$. *[Q1623]*
- Lime ($Ca(OH)_2$) + dilute acid → salt + water; CO_2 nahi banta. *[Q1615]*
- Baking soda ($NaHCO_3$), limestone ($CaCO_3$), marble ($CaCO_3$) — dilute acid se CO_2 dete hain. *[Q1615]*
- Glacial acetic acid: 100% acetic acid free of water (<1%); colorless, corrosive; 16.7°C pe solidify. *[Q1621]*
- NaOH (Caustic Soda): Basic nature OH^- ions ki wajah se — H^+ ions se nahi. *[Q1687]*
- H_2SO_4: Petroleum refining mein "sweetening/desulfurization" catalyst — sulfur hataane ke liye. *[Q1687]*
- Aqua regia: HNO_3 : HCl = 1:3 — gold dissolve karne ke liye.
  - Aur dissolve hote hain: Pd, Ga, Ti. *[Q1633]*
- Brine electrolysis (Chlor-alkali): NaCl(aq) → NaOH + H_2 + Cl_2. *[Q1680, Q1004 from previous chapter]*
- Dry cell: Energy stored as chemical energy. *[Q1679]*
- Antacids: $Al(OH)_3$, $MgCO_3$, $Mg_2O_8Si_3$, $Mg(OH)_2$, $CaCO_3$, $NaHCO_3$ — excess stomach acid neutralize. *[Q1353, Q1466]*
- Soap molecule: Hydrophilic head (water-soluble) + Hydrophobic tail (oil-soluble). *[Q1602]*
- Saponification: Soap banane ki process; long-chain fatty acids ke K/Na salts. *[Q1602]*

- ⚠️ Trap: Lime dilute acid se CO_2 nahi deta. *[Q1615]*
- ⚠️ Trap: NaOH basic nature = OH^- ions se. *[Q1687]*
- ⚠️ Trap: Soap = hydrophilic head + hydrophobic tail — donon same nahi. *[Q1602]*

7. Important Compounds & Their Reactions

- Baking soda ($NaHCO_3$) heating: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (above 80°C).
  - Na_2CO_3 recrystallization → Washing soda ($Na_2CO_3 \cdot 10H_2O$). *[Q1570]*
- Bleaching powder {$CaOCl_2$}: Dry Cl_2 + dry $Ca(OH)_2$ — oxidising agent, disinfectant. *[Q1446, Q1453, Q1468]*
- Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$): Gypsum ko 373 K par heat karke.
  - Setting: $CaSO_4 \cdot \frac{1}{2}H_2O + H_2O \rightarrow CaSO_4 \cdot 2H_2O$; mass expand + heat liberate. *[Q1614, Q1355]*
- White washing drying: $Ca(OH)_2 + CO_2$ (air) $\rightarrow CaCO_3 + H_2O$ + heat = carbonation. *[Q1612, Q1275]*
- Quicklime (CaO) + H_2O → Slaked lime ($Ca(OH)_2$) — exothermic. *[Q1551]*
- CaO mein Ca:O mass ratio = 40:16 = 5:2. *[Q1663]*
- Slaked lime ($Ca(OH)_2$): Soil acidity correct karne ke liye — pH 1 unit badhane ke liye ~1500 kg/hectare. *[Q1667]*
- $CaCl_2$: Hygroscopic — moist gas dry karne ke liye. *[Q1660]*
- Heavy water (D_2O): Density 11% zyada; neutron moderator in nuclear reactors. *[Q1583]*
- RDX: Royal Demolition Explosive; nitramine; white crystalline; water-insoluble. *[Q1584]*
- Poly-oxime gel: Oct 2018, Indian scientists (InStem) — farmers ko toxic pesticides se bachane ke liye. *[Q1585]*
- Nessler's reagent (K_2HgI_4): Ammonia detect karne ke liye. *[Q1599]*
- Liquor ammonia (NH_4OH): Oil/grease stains hataane ke liye. *[Q1599]*
- Copper sulphate ($CuSO_4$): Algae control ke liye generally used chemical — fungicide, algaecide, root killer, antimicrobial bhi. *[Q1626]*
- Potassium sulphate (K_2SO_4): Elements = K (Potassium), S (Sulphur), O (Oxygen) — sabhi options correct. *[Q1640]*
- Water of crystallization:
  - Baking soda ($NaHCO_3$): NO water of crystallization. *[Q1634]*
  - Washing soda ($Na_2CO_3 \cdot 10H_2O$): 10 molecules. *[Q1634]*
  - Blue vitriol ($CuSO_4 \cdot 5H_2O$): 5 molecules. *[Q1634]*
  - Gypsum ($CaSO_4 \cdot 2H_2O$): 2 molecules. *[Q1634]*
- ⚠️ Trap: Baking soda garam karne se Na_2CO_3 banta hai — NaOH ya Na_2O nahi. *[Q1575]*
- ⚠️ Trap: POP setting mein mass expand + heat liberate dono hote hain. *[Q1614]*
- ⚠️ Trap: Baking soda mein NO water of crystallization. *[Q1634]*

8. Metals, Rust, Corrosion & Properties

- Rust = Hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) — reddish-brown; iron + O_2 + water. *[Q1630, Q1546]*
- Ferric chloride ($FeCl_3$): Orange to brown-black; slightly soluble. *[Q1630]*
- Ferric bromide ($FeBr_3$): Red-brown; Lewis acid catalyst. *[Q1630]*
- Ferric sulphide (Fe_2S_3): Black powder; degrades at ambient temp. *[Q1630]*
- Gold (Au): Highest density among common metals; 24k = pure; dissolves in aqua regia. *[Q1647]*
- Osmium (Os, Z=76): Densest element. Hydrogen (H, Z=1): Lowest density. *[Q1647, Q1598]*
- Mercury (Hg, Z=80): Quicksilver; only metallic element liquid at RT. *[Q1647, Q1533]*

- Copper (Cu, Z=29): First metal used by man. *[Q1647]*
- Iron (Fe, Z=26): Wrought iron = purest; Pig iron = most impure. *[Q1647]*
- Copper valency: Variable — +1 in Cu_2O (cuprous), +2 in $CuO/CuSO_4$ (cupric). *[Q1655]*
- Zinc: Bluish-grey metal. SO_2: Sharp pungent smell. PbI_2: Bright yellow ("golden rain"). *[Q1656]*
- Transition metals: Paramagnetic (unpaired electrons); d-block config $(n-1)d^{1-10}\ ns^{1-2}$; variable valency, colored ions, catalytic. *[Q1641]*
- Electrode materials: Copper, graphite, titanium, brass, silver, platinum. *[Q1668]*
- Silver plating: Silver electrode use hota hai. *[Q1668]*
- Galvanization: Iron/steel pe zinc coating — corrosion prevention. *[Q1500]*
- ⚠ Trap: Rust = $Fe_2O_3 \cdot xH_2O$ — ferric chloride/bromide/sulphide nahi. *[Q1630]*
- ⚠ Trap: Cuprous = Cu^{+1}; Cupric = Cu^{+2}. *[Q1655]*
- ⚠ Trap: Densest element = Osmium — platinum/steel/copper nahi. *[Q1598, Q1647]*

9. Separation Techniques

- Magnetic separation: Iron pins from sand. *[Q1620]*
- Winnowing: Husk from wheat grains — lighter husk air mein ur jaati hai. *[Q1625]*
- Centrifugation: Butter from cream — rapid spinning. *[Q1646]*
- Froth floatation: Ore separation based on wetting properties; mustard oil as frother. *[Q1644]*
- Filtration: Insoluble impurities from liquid — sand + water. *[Q1625, Q1437]*
- Leaching: Selective dissolution of metals from ores. *[Q1644]*
- Evaporation: Liquid → vapour — salt from seawater. *[Q1425, Q1580]*
- Sublimation: Solid → gas — Iodine, Camphor, NH_4Cl, Naphthalene, Dry ice. *[Q1625, Q1652, Q1681]*
- Separating funnel: Immiscible liquids — oil & water. *[Q1429, Q1499]*
- ⚠ Trap: Sodium sulphate sublimable nahi hai. *[Q1652]*

10. Environmental Chemistry

- BOD (Biochemical Oxygen Demand): Dissolved oxygen jo bacteria organic waste decompose karne mein use karte hain — mg/L. *[Q1616]*
- COD (Chemical Oxygen Demand): Oxygen required to oxidize organic + inorganic — ppm. *[Q1616]*
- Carbonaceous demand: Organic matter ke oxidation se. *[Q1616]*
- Ozone depleting substances (ODS): CFCs, HCFCs, halons, methyl bromide, carbon tetrachloride, hydro bromo fluorocarbons.
  - NOT ozone depleting: Sodium silicate. *[Q1636, Q1478]*
- Photochemical smog: Sunlight + NO_x + VOCs. *[Q1608]*
- Sulphurous smog: Sulphur-bearing fossil fuels (coal) se. *[Q1608]*
- Greenhouse gases: CO_2, CH_4, N_2O. *[Q1480, Q1606]*
- GWP (Global Warming Potential): CO_2 ko reference maanke compare karte hain.
  - High-GWP: CFCs, HFCs, HCFCs, PFCs, SF_6. *[Q1606]*
- Natural gas: Anaerobic bacteria se vegetable matter decompose — mainly methane (CH_4). *[Q1607]*
- Conventional energy = Non-renewable: Fossil fuels (coal, oil, natural gas). *[Q1609]*
- Renewable: Sunlight, wind. Inexhaustible: Sunlight. Alternative: Wind, geothermal, hydroelectric, hydrogen.

[Q1609]

- Wind blowing ka cause: Air pressure difference aur Earth ki unequal heating hai — Earth ki rotation akele wind ka cause nahi hai. *[Q1684]*
- Coal composition: Primarily Carbon (C), Hydrogen (H), Oxygen (O) ka complex mixture, saath thoda Nitrogen (N) & Sulphur (S) compounds bhi — "O_2H_2 + half free carbon" wala statement galat hai. *[Q1684]*
- Environment-friendly fuel: Jiske combustion products non-poisonous hon. *[Q1684]*
- Destructive distillation of coal: Air ke bina heat karne se coke banta hai — grey, hard, porous, high carbon.
  - Coke: Steel manufacture & metal extraction. *[Q1638]*
  - Petrol/kerosene/diesel: Crude oil se, coal se nahi. *[Q1638]*
- Biogas: 50–75% CH_4, 25–50% CO_2, traces H_2S & NH_3. CO nahi hota. *[Q1480, Q1686]*
- ⚠ Trap: Conventional energy = Non-renewable — renewable/inexhaustible/alternative nahi. *[Q1609]*
- ⚠ Trap: GWP reference = CO_2 — methane/ozone/NO_2 nahi. *[Q1606]*
- ⚠ Trap: Biogas mein CO nahi hota. *[Q1686]*
- ⚠ Trap: Wind rotation-of-Earth se akele nahi banta; coal mein O_2H_2 + free carbon nahi hota. *[Q1684]*

11. Gases, Fuels & Energy

- Calorific value order: H_2 (150,000 kJ/kg) > LPG (55,000) > Methane/CNG (50,000). *[Q1603]*
- Hydrogen sabse zyada calorific value rakhta hai. *[Q1603]*
- Calorific value: Heat produced by complete combustion of unit mass — unit kJ/kg. *[Q1643]*
- Ignition temperature: Minimum temp for fuel to catch fire. *[Q1643]*
- Volatile matter: Components readily burnt in oxygen. *[Q1643]*
- Thermal capacity: Heat to produce unit temp change in unit mass. *[Q1643]*
- Cryogenic engine: Liquid O_2 + liquid H_2 as fuel.
  - RL-10 (world's first, 1963, USA); CE-20 (India, 5 June 2017). *[Q1589]*
- LPG components: Propane, butane, propylene, butylene, isobutane. Main = Butane. *[Q1398, Q1498]*
- Electric bulbs: Filled with N_2 + Ar — tungsten filament oxidation rokne ke liye. *[Q1493]*
- CO: Odourless, colorless; incomplete combustion se; oxygen displace karke poisoning. *[Q1493]*
- NPK fertilizer: N (leaf growth), P (root/flower/fruit), K (protein synthesis). *[Q1590]*
- Ammonia (NH_3): N:H mass ratio = 14:3. *[Q1593]*
- Helium (He): 2nd lightest; balloons, medicine, arc welding, cryogenic research. *[Q1481]*
- Nitrogen (N_2): Food preservation — displaces oxygen; atmosphere ~78%; Daniel Rutherford (1772). *[Q1501, Q1510]*
- Oxygen essential for burning — candle glass jar mein cover karne se bujh jaati hai. *[Q1592]*
- Dry ice (Solid CO_2): Sublimes directly to gas — vaccine cooling, food freezing, special effects fog. *[Q1658, Q1648, Q1558]*
- Nuclear energy: Atoms se obtained energy ko nuclear energy kehte hain — Uranium fission se heat, steam banta hai jo turbine generator se electricity generate karta hai. *[Q1657]*
- ⚠ Trap: Calorific value sabse zyada Hydrogen ka hai — LPG/methane/natural gas nahi. *[Q1603]*
- ⚠ Trap: Nitrogen food preservation ke liye — oxygen/hydrogen/chlorine nahi. *[Q1501]*

12. Organic Chemistry, Fats & Nutrition

- Wohler ne disprove kiya "organic compounds sirf living system mein bante hain". *[Q1567]*

- Organic compound: Carbon covalently bound to other atoms (C-C, C-H). *[Q1567]*
- Propanoic acid (CH_3CH_2COOH): Suffix = "oic acid"; antifungal food agent. *[Q1568]*
- Ethane (C_2H_6): Total 7 covalent bonds — 1 C-C + 6 C-H. *[Q1540]*
- Amino acids:
  - Cysteine: Sulphur-containing, non-essential.
  - Methionine: Sulphur-containing, essential.
  - Tryptophan: Aromatic, essential.
  - Histidine: Basic. Serine: Non-essential. *[Q1594]*
- Saturated fats: Solid at RT — single bonds, hydrogen saturated; butter, palm/coconut oil, cheese, meat. *[Q1639]*
- Unsaturated fats: Liquid at RT — double bonds; healthier. *[Q1639]*
- Monounsaturated fats: Liquid at RT, harden when chilled. *[Q1639]*
- Polyunsaturated fats: Omega-3, Omega-6 — essential for brain function. *[Q1639]*
- Fermentation: Glucose ka anaerobic breakdown — enzymes se chemical change. *[Q1627]*
- Butyric acid: Rancid butter ka foul smell; dairy products. *[Q1487]*
- Stearic acid: Animal & plant fats; soap preparation. *[Q1487]*
- Acetic acid (CH_3COOH): Vinegar; pickled vegetables, sauces. *[Q1505]*
- Rancidity: Fats/oils ka oxidation/hydrolysis; airtight containers/N_2 flush se roka jaata hai. *[Q1390, Q1487]*
- Lactose ($C_{12}H_{22}O_{11}$): Milk sugar; disaccharide. Glucose/Galactose/Fructose = $C_6H_{12}O_6$. *[Q1557]*

13. Polymers, Fibres & Plastics

- Natural fibres: Cotton, silk, wool, linen, jute. *[Q1591, Q1595]*
- Synthetic fibres: Nylon, polyester, acrylic, aramid, rayon (regenerated cellulose). *[Q1591, Q1595]*
- PVC (Polyvinyl chloride): Monomer = vinyl chloride; raincoats, handbags, water pipes, floor tiles. *[Q1483, Q1665]*
- Bakelite: Phenol + formaldehyde; switches, sockets. *[Q1490]*
- Teflon (PTFE): Non-stick coating. *[Q1492]*
- PET: Bottles, utensils. *[Q1488]*
- Nylon: Backpacks, flak jackets, athletic shoes, umbrellas, ropes. *[Q1488]*
- Acrylic: Sweaters, fleece, socks. *[Q1488]*
- Polystyrene: Insulator, wrapping, toys, radio/TV cabinets. *[Q1483, Q1665]*
- Polypropene: Ropes, toys, pipes, fibres. *[Q1665]*
- Polymer definition: Large molecules, high molecular mass (10^3–10^7 u). *[Q1665]*
- Firemen uniforms: Melamine plastic coating — flame resistant. *[Q1596]*
- Glyptal: Paints, metal coating, lacquers. *[Q1483]*

14. Health, Safety & Everyday Applications

- WHO salt recommendation: < 5 g per day — hypertension, heart disease, stroke prevention.
  - 2025 tak global salt intake 30% reduce karne ka target.
  - Iodine: Adults 150 mcg, Pregnant 220 mcg.
  - Magnesium: 19-30 M 400 mg / F 310 mg; 31+ M 420 mg / F 320 mg.
  - Calcium: 14-18 age 1300 mg; 19+ 1000 mg. *[Q1581]*

- Cobalt-60: Cancer treatment — gamma rays. *[Q1597]*
- Iodine-131: Thyroid cancer treatment. *[Q1597]*
- Uranium-235: Nuclear reactors. *[Q1597]*
- Chocolates mein heavy metals: Nickel (Ni-28), Cadmium (Cd-48), Lead (Pb-82) — nickel sabse toxic. *[Q1604]*
- Pickle preservatives: Salt, sodium benzoate, sodium metabisulphite. Pepper preservative nahi — sirf flavour. *[Q1600]*
- Foxing: Old books ke pages ka browning. *[Q1582]*
- Water hardness:
  - Soft: 0–60 mg/L as $CaCO_3$.
  - Moderately hard: 61–120 mg/L. Hard: 121–180 mg/L. Very hard: >180 mg/L. *[Q1613]*
- Toothpaste: Basic (pH > 7) — mouth bacteria acid neutralize; pH 5.5 se neeche tooth decay. *[Q1463]*
- Osmotic concentration: Solute concentration — osmoles per litre. *[Q1635]*
- Osmotic pressure: Minimum pressure to prevent inward flow of pure solvent across semipermeable membrane. *[Q1635]*
- Turgor pressure: Fluid pressure pressing cell membrane against cell wall. *[Q1635]*
- Wall pressure: Cell wall pressure against turgor pressure. *[Q1635]*
- Water movement between cells: Osmotic concentration pe depend karta hai. *[Q1635]*
- Surface tension: Liquid surface resists external force. *[Q1586]*
- Cohesive force: Same substance ke molecules ke beech attraction. *[Q1586]*
- Adhesive force: Different substances ke molecules ke beech attraction. *[Q1586]*
- Capillarity: Liquid rise/fall in capillary tube due to surface tension. *[Q1586]*
- India's first synthetic drug — Methaqualone: CSIR (Council of Scientific and Industrial Research, est. 26 Sept 1942, HQ New Delhi — India ka largest R&D organization) dwara develop kiya gaya synthetic barbiturate-like CNS depressant drug. *[Q1601]*
- ⚠️ Trap: Pepper pickle mein preservative nahi hai. *[Q1600]*
- ⚠️ Trap: "Size" chemical reaction ka indicator nahi hai. *[Q1208]*

15. Laboratory Equipment & Techniques

- China dish: Round-bottomed crucible-shaped porcelain; chemical reactions; delicate but strong. *[Q1565]*
- Flame zones:
  - Innermost (Dark/Black): Incomplete combustion, lack of O_2 — unburnt carbon particles.
  - Middle (Yellow/Luminous): Sufficient O_2, glowing heated carbon particles.
  - Outermost (Blue): Complete combustion, ample O_2. *[Q1588]*
- Diffusion: Particles move high concentration → low concentration.
  - Graham's Law: Rate ∝ 1/√(density/molecular weight) — at constant T & P.
  - Diffusion rate: Gas > Liquid > Solid.
  - Diffusion depends on: Temperature, pressure, density — NOT volume. *[Q1649]*
- Electrolysis of water: Electricity se water → $H_2 + O_2$.
  - Ionisation = weak electrolyte self-ionisation.
  - Hydrolysis = water molecule breaks chemical bonds.
  - Atomisation = high pressure water spray on molten metal → fine powder. *[Q1579]*

16. ⚠ Traps / Key Differences Summary (Full)

- ⚠ Trap: Silicon long-chain compounds hyper-reactive hain — stable nahi. *[Q1561]*
- ⚠ Trap: N_2 mein 6 shared electrons hain — 4 ya 8 nahi. *[Q1562]*
- ⚠ Trap: CO_2 mein har oxygen carbon ke saath 2 electrons share karti hai. *[Q1564]*
- ⚠ Trap: $MgCl_2$ ionic hai — covalent nahi. *[Q1566]*
- ⚠ Trap: Ionic compounds solid hain kyunki strong inter-ionic attraction hai — CaO (2850K) sabse high melting/boiling point. *[Q1572, Q1563, Q1573, Q1578, Q1661]*
- ⚠ Trap: NaOH ionic + covalent dono hai — pure covalent nahi. *[Q1576]*
- ⚠ Trap: Baking soda garam karne se Na_2CO_3 banta hai — NaOH ya Na_2O nahi. *[Q1575]*
- ⚠ Trap: Electrolysis of water ≠ Hydrolysis ≠ Ionisation ≠ Atomisation. *[Q1579]*
- ⚠ Trap: Sublimation = Solid → Gas; Deposition = Gas → Solid. *[Q1580]*
- ⚠ Trap: Heavy water ka density sirf 11% zyada hota hai — 3 times nahi. *[Q1583]*
- ⚠ Trap: RDX = explosive — pesticide/current meter nahi. *[Q1584]*
- ⚠ Trap: Flame ka innermost part unburnt carbon particles ki wajah se black. *[Q1588]*
- ⚠ Trap: Liquid Nitrogen ka boiling point -196°C hai. *[Q1587]*
- ⚠ Trap: Calorific value sabse zyada Hydrogen ka hai. *[Q1603]*
- ⚠ Trap: Fermentation chemical change hai. *[Q1605]*
- ⚠ Trap: GWP reference = CO_2. *[Q1606]*
- ⚠ Trap: Conventional energy = Non-renewable. *[Q1609]*
- ⚠ Trap: Soap = hydrophilic head + hydrophobic tail. *[Q1602]*
- ⚠ Trap: Pepper pickle mein preservative nahi. *[Q1600]*
- ⚠ Trap: Lime dilute acid se CO_2 nahi deta. *[Q1615]*
- ⚠ Trap: POP setting mein mass expand + heat liberate dono hote hain. *[Q1614]*
- ⚠ Trap: Glacial acetic acid = 100% acetic acid — frozen nahi, water-free. *[Q1621]*
- ⚠ Trap: Alkali = base jo water mein dissolve hoti hai. *[Q1623]*
- ⚠ Trap: Electron add karne se anion banta hai. *[Q1624]*
- ⚠ Trap: NaCl mein cation = Na^+ — Cl^+ nahi. *[Q1628]*
- ⚠ Trap: Iodine + potato = blue-black. *[Q1631]*
- ⚠ Trap: Baking soda mein NO water of crystallization. *[Q1634]*
- ⚠ Trap: Diffusion rate volume pe depend nahi karta. *[Q1649]*
- ⚠ Trap: AgCl sunlight mein grey hota hai — brown nahi; AgCl banta hai Ag^+ + Cl^- se. *[Q1650, Q1685]*
- ⚠ Trap: Carbon sabse zyada melting/boiling point — Li/P/Ar nahi. *[Q1651]*
- ⚠ Trap: Sodium sulphate sublimable nahi hai. *[Q1652]*
- ⚠ Trap: Methane = pure substance; milk/sugar paste/air = mixture. *[Q1653]*
- ⚠ Trap: Intermolecular force solid > liquid > gas. *[Q1654]*
- ⚠ Trap: Cuprous = Cu^{+1}; Cupric = Cu^{+2}. *[Q1655]*
- ⚠ Trap: Slaked lime soil quality improve karta hai — manure/vinegar nahi. *[Q1667]*
- ⚠ Trap: Silver = electrode material; Zn/Al/Ca nahi. *[Q1668]*
- ⚠ Trap: 1 mole mein hamesha 6.022×10^{23} particles — atomic mass pe depend nahi. *[Q1673]*
- ⚠ Trap: "Atom of a compound" — galat statement. *[Q1675]*
- ⚠ Trap: NaOH basic nature = OH^- ions se; H^+ se nahi. *[Q1687]*
- ⚠ Trap: Biogas mein CO nahi hota. *[Q1686]*
- ⚠ Trap: Solids ka definite shape hota hai. *[Q1676]*

- ⚠️ Trap: Particles stationary nahi — continuous motion mein hain. *[Q1677]*
- ⚠️ Trap: Wind rotation-of-Earth se akele nahi banta; coal mein O_2H_2 + free carbon nahi hota. *[Q1684]*
- ⚠️ Trap: India ka first synthetic drug Methaqualone tha (CSIR). *[Q1601]*
- ⚠️ Trap: Evaporation rate surface area aur temperature dono se badhti hai. *[Q1642]*
- ⚠️ Trap: Nuclear energy = atoms se obtained energy. *[Q1657]*
- ⚠️ Trap: Copper sulphate algae control ke liye use hota hai. *[Q1626]*